Distributed Adaptive Flooding for Reducing Redundancy in Wireless Multi-Hop Networks

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***Abstract*—Flooding is a fundamental communication primitive in wireless multi-hop networks, but its simplicity often leads to excessive redundancy and communication overhead, especially in dense environments. In this work, we propose a distributed adaptive flooding protocol that reduces redundant transmissions through local decision-making. Instead of forwarding messages immediately upon first reception, each node delays its decision within a short time window and uses duplicate receptions as a signal of local redundancy. The forwarding probability is further adjusted based on local node degree, allowing nodes in dense regions to suppress transmissions more aggressively.**

We evaluate the proposed method against baseline flooding under varying network densities using simulation. Results show that the adaptive protocol significantly reduces forwarding overhead and redundant receptions in dense networks while maintaining full delivery. In medium-density networks, a tradeoff emerges between efficiency and reliability, with moderate reductions in delivery ratio. In sparse networks, performance is dominated by network connectivity rather than forwarding strategy, and both methods exhibit limited reachability.

These findings demonstrate that local adaptation can improve the efficiency of distributed broadcast, but its effectiveness depends strongly on network density and connectivity. The results highlight the importance of balancing redundancy suppression with robustness in the design of distributed communication protocols.

***Index Terms*—wireless networks, flooding, distributed protocols, broadcast suppression, adaptive forwarding**

I. Introduction

Flooding is one of the most fundamental dissemination mechanisms in distributed wireless networks. Its appeal lies in its simplicity: when a node receives a message for the first time, it forwards that message to its neighbors, and the process continues until the message reaches the network boundary. This design requires no global coordination, no route construction, and no centralized controller, making it attractive for highly dynamic or infrastructure-free environments. However, this simplicity comes at a cost. In dense networks, flooding generates substantial redundant transmissions, repeated receptions, and unnecessary contention for the wireless medium. As a result, the broadcast primitive that initially appears robust can become highly inefficient as network density increases.

The central observation motivating this work is that redundancy is not uniformly harmful, but it is also not uniformly useful. In a sparse or weakly connected network, aggressive suppression of retransmissions may cause the dissemination process to die out prematurely. In contrast, in a dense network, many retransmissions contribute little new information and mostly duplicate already available paths. This suggests that the forwarding decision should not be fixed a priori. Instead, a node should adapt its behavior according to local evidence about how widely the message is already propagating in its neighborhood.

In this project, we study a distributed adaptive flooding protocol for wireless multi-hop networks. The protocol remains fully decentralized: each node makes forwarding decisions using only local observations, namely the number of duplicate receptions it has seen for a message and its local neighborhood degree. Rather than forwarding immediately upon the first reception, a node buffers the message for a short decision window and then decides whether to retransmit based on a probability that decreases as redundant receptions accumulate. This design is intended to preserve the essential reachability of flooding while reducing unnecessary transmissions in dense regions.

We evaluate the protocol in a lightweight wireless network simulator under three connectivity regimes: dense, medium, and sparse. The results show that the adaptive scheme can substantially reduce forwarding overhead in dense networks while maintaining full delivery in the tested settings. In medium-density networks, a tradeoff emerges between efficiency and reliability, and in sparse networks performance becomes dominated by network connectivity rather than forwarding policy. Taken together, these results suggest that local adaptation can improve the efficiency of distributed broadcast, but only when it is tuned with respect to the underlying topology.

II. Related Work

Flooding is a fundamental broadcast mechanism in wireless multi-hop networks and has been extensively studied due to its simplicity and robustness. In its basic form, flooding requires each node to retransmit a message upon first reception, ensuring high reachability in connected networks. However, this approach often leads to excessive redundancy, commonly referred to as the broadcast storm problem, especially in dense networks [1].

To address this issue, a variety of optimized flooding techniques have been proposed. Probabilistic flooding schemes reduce redundancy by allowing nodes to forward messages with a fixed probability [2]. While simple to implement, these approaches often require careful tuning and may suffer from reduced reliability in sparse or heterogeneous networks. Other methods, such as counter-based and distance-based schemes, use local metrics to suppress transmissions when sufficient redundancy is detected. These techniques attempt to balance efficiency and coverage but typically rely on heuristic thresholds that may not generalize well across different network conditions [3].

More advanced approaches introduce adaptive mechanisms that adjust forwarding behavior based on network characteristics. Some protocols incorporate neighborhood information or local density estimation to guide retransmission decisions. However, many of these methods either require additional coordination overhead or assume partial knowledge of the network structure, which may not be available in fully distributed settings.

In contrast, our work focuses on a fully distributed adaptive flooding scheme that relies solely on locally observable information, specifically duplicate message receptions and node degree. By introducing a delayed decision mechanism, nodes can infer local redundancy before committing to a retransmission. This approach maintains the simplicity and decentralization of flooding while enabling dynamic adaptation to network density, without requiring global knowledge or explicit coordination.

III. Methodology

We consider a multi-hop wireless network in which nodes communicate through broadcast transmissions over a fixed transmission range. The baseline flooding protocol follows the standard rule: a node forwards a message the first time it sees it and ignores subsequent duplicates. While this rule guarantees broad propagation in connected networks, it also produces significant redundant traffic when many nodes can hear the same message through multiple paths.

Our adaptive protocol modifies only the forwarding policy; it does not introduce a centralized coordinator or any global view of the topology. Each node operates using local state associated with each message origin. Specifically, a node tracks how many times it has received a given flooded message, whether it has already forwarded that message, and the local degree of the node as determined by its immediate neighbors. The key idea is to treat duplicate receptions as a signal of local redundancy. If the same message arrives repeatedly within a short interval, the node infers that the surrounding region is already well covered and can safely suppress additional retransmissions with higher probability.

To make this inference operational, the protocol uses a delayed-decision mechanism. When a node receives a flood message for the first time, it does not forward immediately. Instead, it stores the message in a temporary buffer and schedules a forwarding decision after a short decision window. During this window, the node may receive duplicates of the same message from other neighbors. These duplicates increase the local reception count and therefore reduce the probability of forwarding. The delayed decision allows the node to observe a small amount of local redundancy before committing to a retransmission.

The forwarding probability is designed to depend on two local signals. First, nodes with more neighbors are expected to experience more redundant coverage, so their base forwarding probability is lower than that of nodes with fewer neighbors. Second, each additional duplicate reception further decreases the forwarding probability. In effect, sparse local evidence of redundancy leads to aggressive forwarding, while strong evidence of overlap leads to suppression. To avoid premature extinction of the broadcast, the protocol includes a probability floor and assigns a high forwarding preference to the earliest receptions. This preserves reachability in connected regions while still filtering unnecessary retransmissions in dense regions.

Algorithmically, the process is simple. The source node injects a message into the network. When a node receives the message for the first time, it records the event, starts a timer, and waits. If, by the end of the timer, the message has arrived only once or twice, the node is likely to forward it. If many duplicates arrive during the waiting period, the node is more likely to suppress transmission. Once a node decides to forward, it marks the message as forwarded and broadcasts it to its neighbors. Because each node uses only local message history and neighbor information, the protocol remains distributed and scalable.

We evaluate the method by comparing it against baseline flooding under identical network instances and identical random seeds. For each scenario, we measure delivery ratio, total forwarding count, and total receptions. Delivery ratio captures reliability, while total forwarding count and total receptions capture communication overhead and redundancy. This setup allows us to study the efficiency-reliability tradeoff introduced by local adaptation and to characterize the conditions under which the proposed protocol is beneficial.

IV. Experimental Setup

TABLE I
Simulation Parameters for Different Network Densities

Scenario	n	Area Size	TX Range
Dense	30	10	2.2
Medium	30	15	1.8
Sparse	30	20	1.5

We evaluate the proposed adaptive flooding protocol under three representative network regimes: dense, medium, and sparse. Each configuration is tested over 20 random seeds to ensure statistical reliability. The simulator uses a fixed transmission range and randomized node placement within a bounded square area.

The performance metrics are:

- **Delivery ratio**: fraction of nodes that receive the flooded message.
- **Total forwards**: total number of transmissions triggered by the protocol.
- **Total receptions**: total number of message receptions across the network.

V. Results

We compare the adaptive scheme against baseline flooding using the metrics above. The results are summarized across dense, medium, and sparse network settings.

A. Dense Networks

In dense network configurations, baseline flooding achieves perfect delivery but incurs substantial redundancy. As shown in Fig. 1 and Fig. 2, the adaptive flooding protocol maintains full delivery while reducing communication overhead.

Specifically, the adaptive scheme reduces total forwarding count from approximately 30 to 26.65 on average, and total receptions from approximately 340 to 300. This corresponds to a reduction of around 10–15% in redundant transmissions.

This result confirms the central hypothesis of this work: in dense networks, duplicate message receptions provide strong local evidence of redundancy, allowing nodes to safely suppress unnecessary retransmissions without sacrificing reachability.

B. Medium-Density Networks

In medium-density networks, the behavior of the adaptive protocol changes significantly. As shown in Fig. 1, both baseline and adaptive flooding achieve less than perfect delivery, with baseline reaching approximately 0.83 and adaptive reaching approximately 0.78 on average.

At the same time, the adaptive scheme continues to reduce communication overhead. The total forwarding count decreases from approximately 24.95 to 22.20, and total receptions from approximately 111 to 99, as shown in Fig. 2 and Fig. 3.

These results highlight a key tradeoff: while adaptive flooding reduces redundancy, it also increases the risk of premature suppression in regions where connectivity is weaker. Unlike dense networks, duplicate receptions are less reliable indicators of global coverage, and local decisions may lead to missed dissemination opportunities.

C. Sparse Networks

In sparse network configurations, both baseline and adaptive flooding perform poorly. The average delivery ratio drops to approximately 0.22 for both protocols, as shown in Fig. 1.

This behavior is not due to the forwarding strategy, but rather to the underlying network topology. In many instances, the network is partially or fully disconnected, meaning that no forwarding policy can ensure complete dissemination.

Under these conditions, the adaptive protocol behaves similarly to baseline flooding, with only minor differences in forwarding counts and receptions. This indicates that when

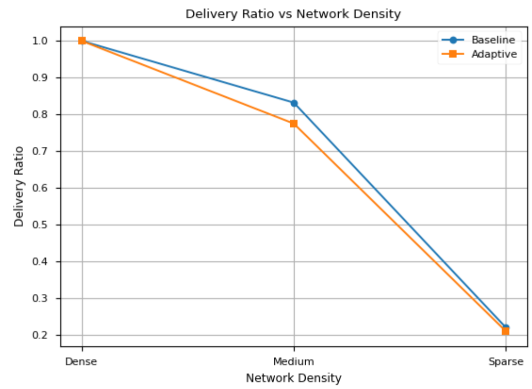

Fig. 1. Delivery ratio versus network density for baseline and adaptive flooding.

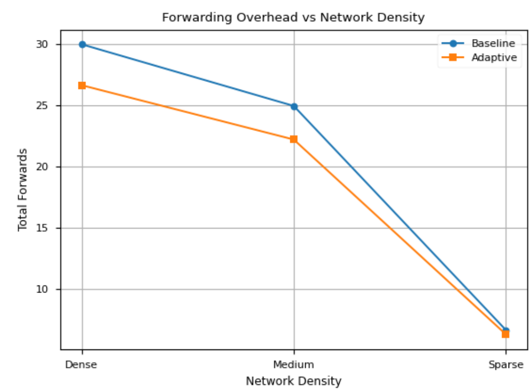

Fig. 2. Total forwarding overhead versus network density for baseline and adaptive flooding.

connectivity is insufficient, protocol-level optimizations have limited impact on overall performance.

D. Summary of Observations

Across all scenarios, three key observations emerge:

- Adaptive flooding significantly reduces redundant transmissions while preserving full delivery in dense networks.
- In medium networks, the protocol continues to reduce overhead, but at the cost of a modest reduction in delivery ratio.
- In sparse networks, both baseline and adaptive flooding are constrained by the underlying topology rather than the forwarding strategy.

These results demonstrate that the benefits of adaptive flooding are highly dependent on network density, and that local suppression mechanisms must be carefully tuned to balance efficiency and reliability.

VI. Discussion

The results highlight that the effectiveness of adaptive flooding is fundamentally tied to network structure, particularly node density and connectivity. While the proposed protocol

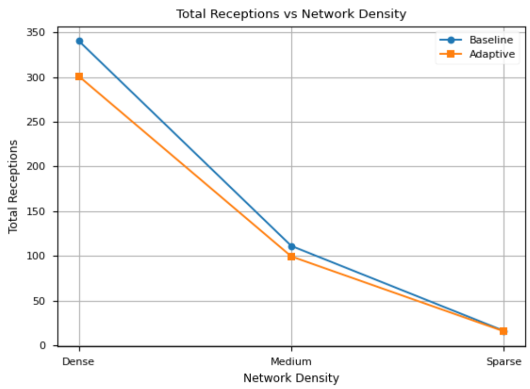

Fig. 3. Total receptions versus network density for baseline and adaptive flooding.

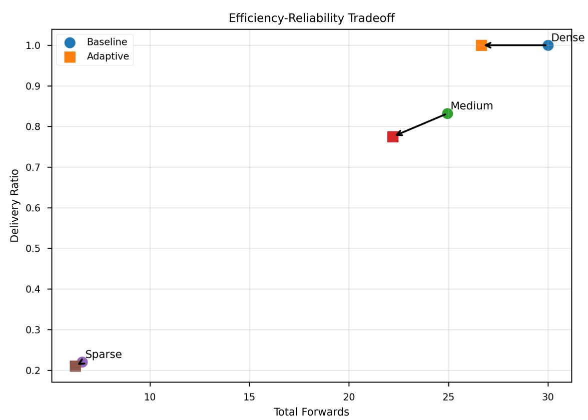

Fig. 4. Efficiency-reliability tradeoff between total forwards and delivery ratio.

consistently reduces redundant transmissions, its impact on delivery varies across different regimes, revealing an inherent tradeoff between efficiency and reliability.

A. Role of Local Redundancy in Distributed Decisions

The core mechanism of the adaptive protocol relies on duplicate message receptions as a local signal of redundancy. In dense networks, this signal is highly informative: multiple receptions within a short time window strongly indicate that the message is already propagating through multiple paths. Under such conditions, suppressing additional retransmissions has minimal impact on reachability but significantly reduces communication overhead.

However, this assumption weakens as network density decreases. In medium-density networks, duplicate receptions become less frequent and less reliable as indicators of global coverage. As a result, nodes may prematurely suppress forwarding, even when additional transmissions are still necessary to ensure full dissemination.

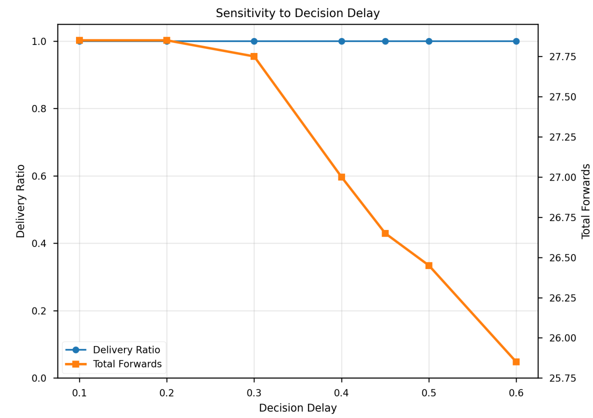

Fig. 5. Sensitivity of delivery ratio and forwarding overhead to the decision delay parameter.

B. Efficiency–Reliability Tradeoff

The results clearly demonstrate a tradeoff between communication efficiency and delivery reliability. The adaptive scheme reduces forwarding overhead across all tested scenarios, but this reduction comes at the cost of decreased robustness in less connected environments.

This tradeoff arises from the fundamentally local nature of the decision process. Each node makes forwarding decisions based only on its immediate observations, without knowledge of the global network state. While this ensures scalability and decentralization, it also limits the ability of the protocol to distinguish between locally redundant transmissions and globally necessary ones.

Importantly, this tradeoff is not unique to the proposed method but is inherent to distributed broadcast mechanisms that rely on local suppression strategies. Any protocol that reduces redundancy without global coordination must balance the risk of suppressing critical transmissions.

C. Impact of Network Connectivity

In sparse networks, both baseline and adaptive flooding exhibit poor performance, with delivery ratios significantly below 1.0. This behavior is primarily determined by the underlying topology rather than the forwarding strategy. In many instances, the network is partially disconnected, meaning that no sequence of retransmissions can reach all nodes.

Under such conditions, protocol-level optimizations have limited impact on overall performance. The similarity in delivery ratios between baseline and adaptive flooding indicates that connectivity, rather than redundancy, is the dominant factor. This observation underscores a key limitation: flooding protocols, whether adaptive or not, cannot overcome fundamental connectivity constraints.

D. Design Implications

These findings suggest several practical implications for the design of distributed broadcast protocols:

- Adaptive suppression is most beneficial in dense environments, where redundancy is high and multiple alternative paths exist.
- In medium-density networks, careful parameter tuning is required to balance efficiency and reliability. Overly aggressive suppression can degrade delivery performance.
- In sparse networks, improving connectivity is more important than optimizing forwarding policies. Mechanisms such as mobility, increased transmission range, or infrastructure support may be required to achieve reliable dissemination.

E. Limitations and Future Work

This study focuses on a simplified wireless model without packet loss, interference, or contention effects. In real-world networks, these factors may amplify both redundancy and transmission failures, potentially changing the balance between efficiency and reliability.

Additionally, the current protocol uses a fixed decision window and heuristic probability function. More advanced designs could dynamically adjust these parameters based on observed network conditions or incorporate additional signals such as temporal patterns or link quality.

Finally, the evaluation considers a single-message dissemination scenario. In practical systems, multiple concurrent broadcasts may interact, introducing additional complexity that warrants further investigation.

VII. Conclusion

In this paper, we presented a distributed adaptive flooding protocol that leverages local observations of duplicate message receptions to reduce communication redundancy in wireless multi-hop networks. By introducing a delayed decision mechanism and adjusting forwarding probabilities based on local density and reception history, the proposed approach maintains the simplicity and decentralization of flooding while improving efficiency.

Through extensive simulations, we showed that the adaptive scheme effectively reduces forwarding overhead in dense networks without sacrificing delivery performance. However, the results also reveal a clear tradeoff between efficiency and reliability in medium-density environments, where overly aggressive suppression can hinder message dissemination. In sparse networks, performance is largely constrained by network connectivity, limiting the impact of protocol-level optimizations.

Overall, this work highlights both the potential and the limitations of local adaptation in distributed broadcast protocols. Future work may explore dynamic parameter tuning, integration with more realistic wireless models, and extensions to multi-message or interference-prone environments.

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